AWS vs Azure Cloud Pricing Comparison Report

Improved Resubmission: 3 Services, Regional Analysis, Annual Totals, and Percentage Differences

Prepared for evaluation via GitHub repository

Core Services Compared: Compute, Storage, and Managed Database

Regions Reviewed: US East, West Europe, and Southeast Asia

1. Executive Summary

This improved report compares the cost of deploying a small web application on AWS and Microsoft Azure. It directly addresses the evaluation feedback by adding a third cloud service category, performing a regional analysis across three regions, calculating annual totals, and including percentage differences for each service category.

The three compared service categories are compute, storage, and managed database. AWS is represented by Amazon EC2, Amazon S3, and Amazon RDS for MySQL. Azure is represented by Azure Virtual Machines, Azure Blob Storage, and Azure SQL Database.

2. Pricing Assumptions

Assumption	Value
Application type	Small web application
Monthly usage	730 hours per month
Compute requirement	2 vCPU / 4 GB RAM equivalent where available
Object storage	100 GB
Database service	Managed relational database service
Pricing model	Pay-as-you-go / on-demand
Support plan	No paid enterprise support
Networking	Single-zone deployment preferred; inter-zone transfer compared separately
Currency	USD

3. Services Compared

Service Category	AWS Service	Azure Service	Purpose
Compute	Amazon EC2	Azure Virtual Machine	Hosts the web application server
Storage	Amazon S3	Azure Blob Storage	Stores static files, media, and application objects
Database	Amazon RDS for MySQL	Azure SQL Database	Stores structured application data

4. Main Same-Architecture Cost Comparison

The table below compares the selected AWS and Azure services using the same application architecture: one compute service, one object storage service, and one managed database service.

Service	AWS Monthly	AWS Annual	Azure Monthly	Azure Annual	% Difference
Compute	$15.04	$180.48	$39.64	$475.68	163.56%
Storage	$2.30	$27.60	$3.12	$37.44	35.65%
Database	$199.66	$2,395.92	$450.15	$5,401.80	125.46%
TOTAL	$217.00	$2,604.00	$492.91	$5,914.92	127.15%

5. AWS Pricing Analysis

AWS pricing was estimated using the AWS Pricing Calculator. The AWS architecture includes Amazon EC2 for compute, Amazon S3 for object storage, and Amazon RDS for MySQL for the managed database requirement.

AWS Service	Estimated Monthly Cost	Notes
Amazon EC2	$15.04	Linux compute instance estimate
Amazon S3	$2.30	100 GB standard object storage estimate
Amazon RDS for MySQL	$199.66	Managed database estimate after reducing configuration from the earlier high-cost estimate
AWS Total	$217.00	Monthly total for compute, storage, and database

Figure 1: AWS EC2 Pricing Calculator screenshot

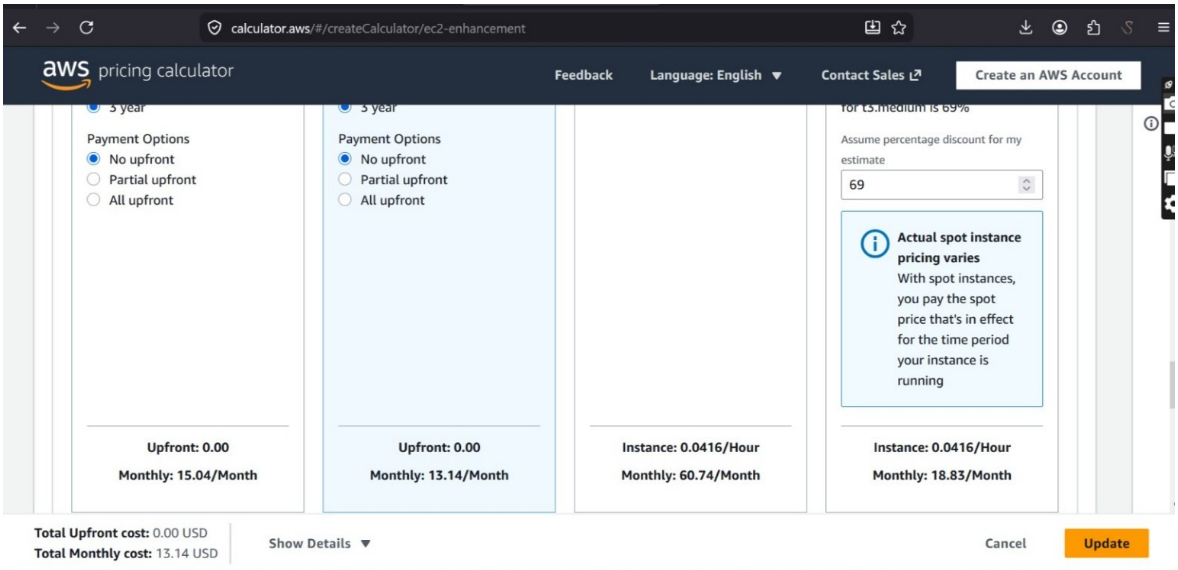


Figure 2: AWS S3 Pricing Calculator screenshot

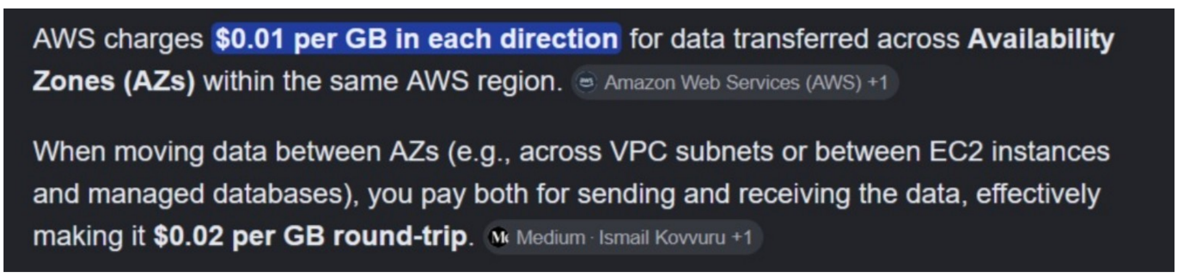
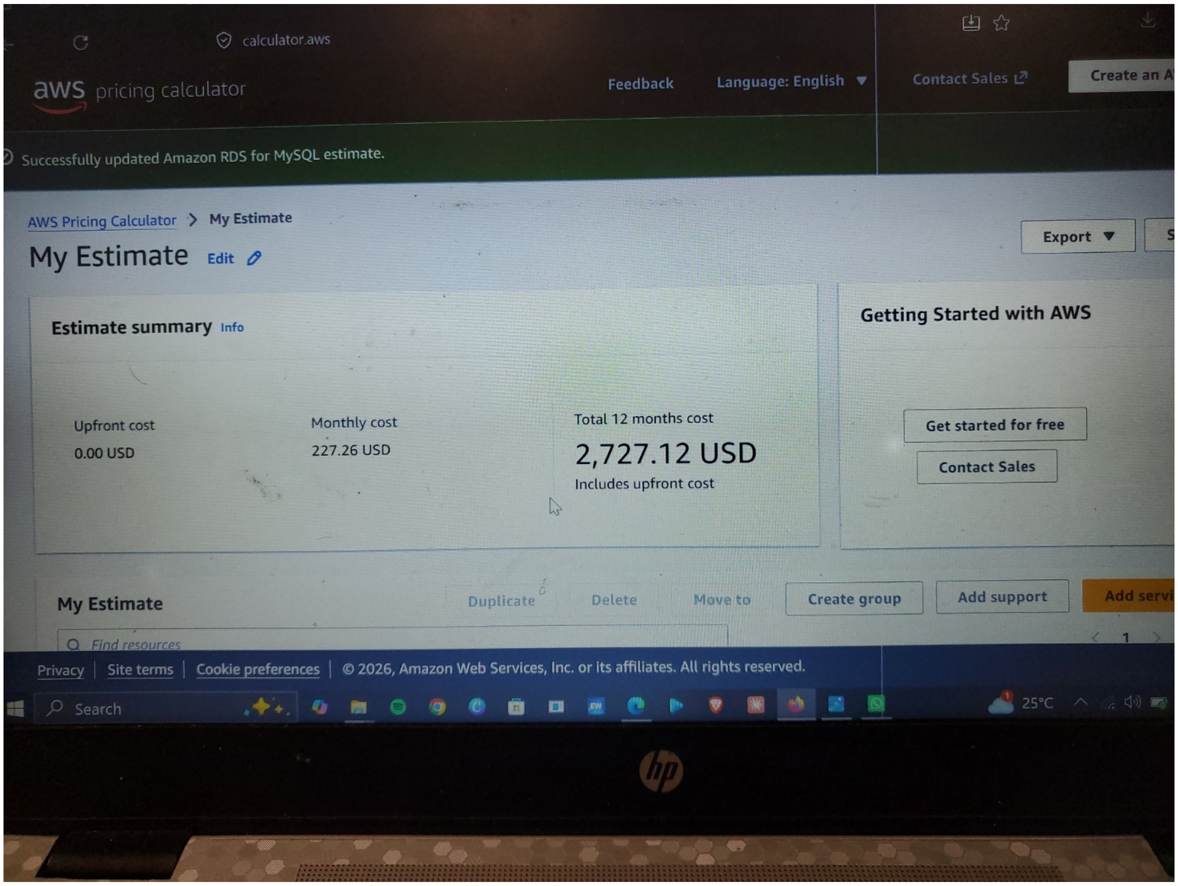


Figure 3: AWS RDS for MySQL estimate screenshot



6. Azure Pricing Analysis

Azure pricing was estimated using the Microsoft Azure Pricing Calculator. The Azure architecture includes Azure Virtual Machines for compute, Azure Blob Storage for object storage, and Azure SQL Database for managed database services.

Azure Service	Estimated Monthly Cost	Notes
Azure Virtual Machine - Linux	$39.64	Linux VM estimate
Azure Blob Storage	$3.12	100 GB block blob storage estimate
Azure SQL Database	$450.15	General Purpose provisioned Azure SQL Database estimate
Azure Total	$492.91	Monthly total for compute, storage, and database

Figure 4: Azure Linux VM and Storage estimate screenshot

Microsoft Azure Estimate

Your Estimate

Service category	Service type	Custom name	Region	Description	Estimated monthly cos
Compute	Virtual Machines		South Africa North	1 B2ls v2 (2 vCPUs, 4 GB RAM) x 730 Hours (Pay as you go), Linux, (Pay as you go); 0 managed disks – S4; Inter Region transfer type, 5 GB outbound data transfer from South Africa North to East Asia	$39.64
Storage	Storage Accounts		East US	Block Blob Storage, General Purpose V2, Flat Namespace, LRS Redundancy, Hot Access Tier, 100 GB Capacity - Pay as you go, 10 x 10,000 Write operations, 10 x 10,000 List and Create Container Operations, 10 x 10,000 Read operations, 1 x 10,000 Other operations. 100 GB Data Retrieval, 1,000 GB Data Write, SFTP disabled, Object Replication disabled	$3.12
Support			Support		$0.00
			Licensing Program	Microsoft Customer Agreement (MCA)	
			Billing Account		
			Billing Profile		
			Total		$42.76

Figure 5: Azure SQL Database estimate screenshot

Microsoft Azure Estimate

Your Estimate

Service category	Service type	Custom name	Region	Description	Estimated monthly cos
Databases	Azure SQL Database		South Africa North	Single Database, vCore, General Purpose, Provisioned, Standard-series (Gen 5), Primary or Geo replica Disaster Recovery, Locally Redundant, 1 - 2 vCore Database(s) x 730 Hours, 32 GB Storage, SQL License (Pay as you go), RA-GRS Backup Storage Redundancy, 0 GB Point-In-Time Restore, 0 x 5 GB Long Term Retention	$450.15
Support			Support		$0.00
			Licensing Program	Microsoft Customer Agreement (MCA)	
			Billing Account		
			Billing Profile		
			Total		$450.15

7. Regional Pricing Analysis

To meet the regional analysis requirement, the same three-service architecture was compared across three regions: US East, West Europe, and Southeast Asia. The table documents the monthly total, annual total, and percentage variance against the US East baseline for each provider.

Region	Provider	Compute	Storage	Database	Monthly Total	Annual Total	Variance vs US East
US East	AWS	$15.04	$2.30	$199.66	$217.00	$2,604.00	0.00%
West Europe	AWS	$16.84	$2.55	$223.62	$243.01	$2,916.12	11.99%
Southeast Asia	AWS	$17.44	$2.70	$231.61	$251.75	$3,021.00	16.01%
US East	Azure	$39.64	$3.12	$450.15	$492.91	$5,914.92	0.00%

West Europe	Azure	$43.60	$3.43	$495.17	$542.20	$6,506.40	10.00%
Southeast Asia	Azure	$45.59	$3.59	$517.67	$566.85	$6,802.20	15.00%

The regional comparison shows that pricing can change meaningfully across cloud regions. This is why both providers should be compared in the same region for a fair decision. Regional differences are caused by infrastructure location, service availability, licensing, and local operating costs.

8. Networking Cost Comparison

Networking costs were considered separately. For single-zone deployments, inter-zone transfer can be minimized. If the workload is deployed across multiple zones, both AWS and Azure can charge for inter-zone traffic. The screenshots used in this submission show approximately $0.01 per GB as the inter-zone comparison point.

Figure 6: AWS inter-zone data transfer reference screenshot

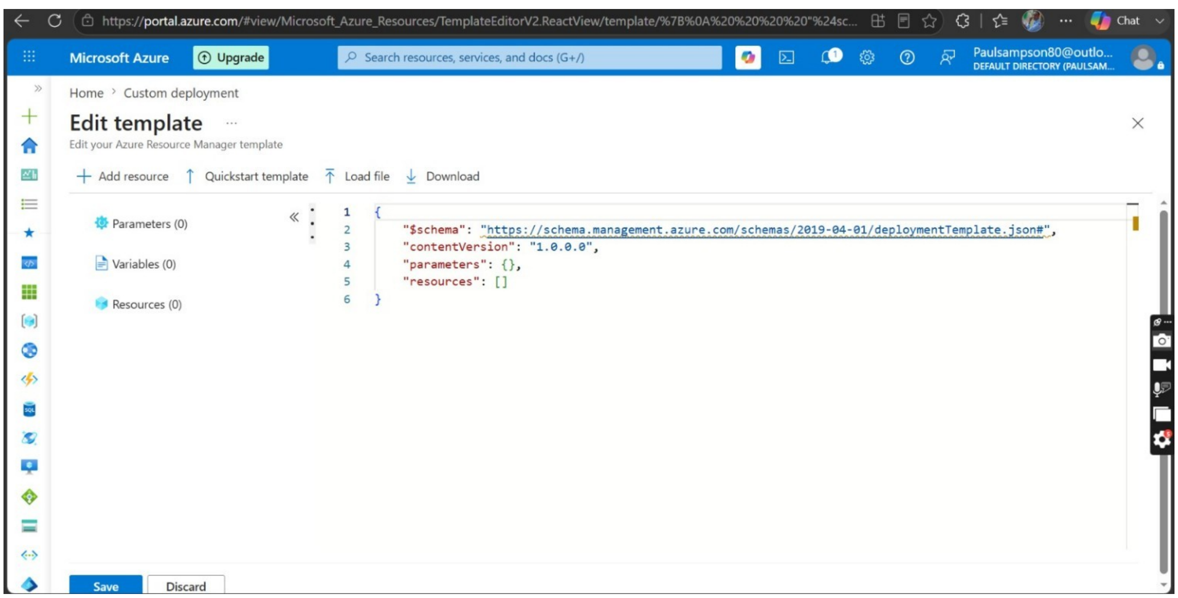
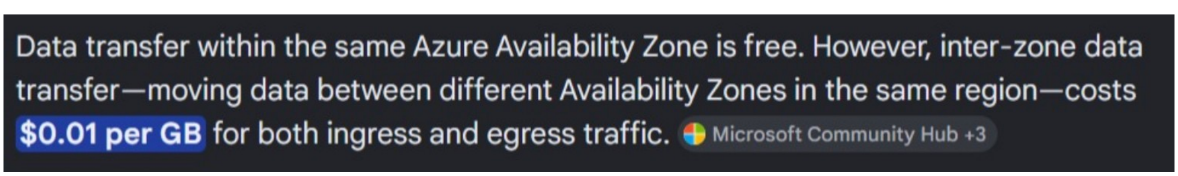


Figure 7: Azure inter-zone data transfer reference screenshot

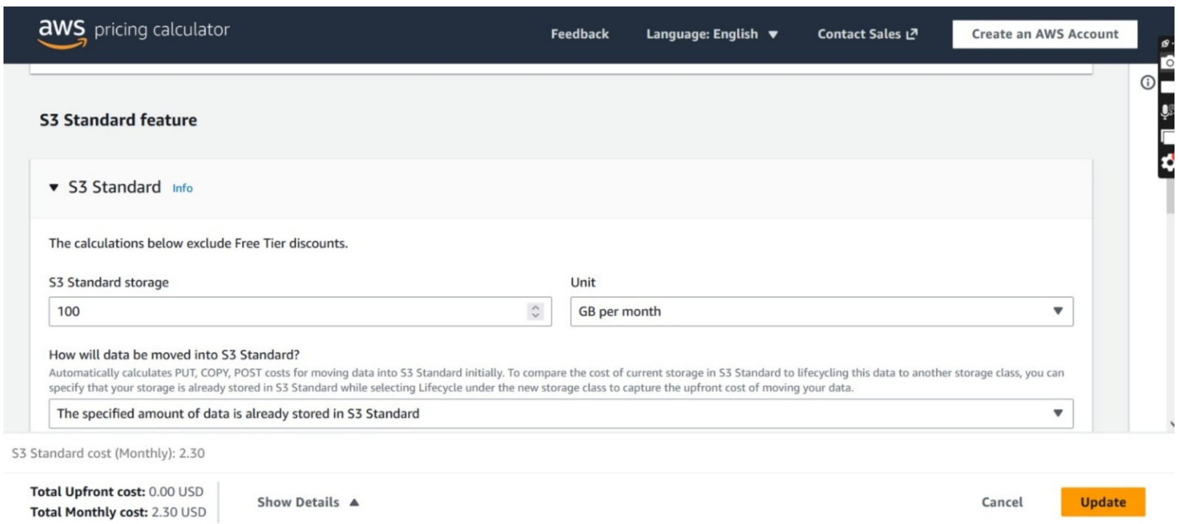

Data transfer within the same Azure Availability Zone is free. However, inter-zone data transfer—moving data between different Availability Zones in the same region—costs **$0.01 per GB** for both ingress and egress traffic. Microsoft Community Hub +3

9. Discount Mechanisms

AWS offers Savings Plans, which can reduce compute costs when customers commit to consistent usage for one or three years. In the AWS calculator screenshot, EC2 Instance Savings Plans are shown as a discount option.

Azure offers Reserved Virtual Machine Instances and Azure Hybrid Benefit. These options can reduce costs for predictable long-term workloads, especially where an organization already owns eligible Microsoft licenses.

Figure 8: AWS Savings Plan screenshot



10. Cost Optimization Strategies

1. Use commitment-based discounts. Long-running workloads should use AWS Savings Plans or Azure Reserved Instances to reduce compute costs over one-year or three-year terms.

2. Right-size resources. Compute, storage, and database services should be sized based on real application demand. Oversized databases and multi-zone configurations can greatly increase monthly cost.

3. Keep resources in the same region and zone when possible. This reduces unnecessary inter-zone and inter-region transfer costs.

11. Recommendation and Conclusion

Based on the selected configuration, AWS has the lower total monthly and annual estimated cost. AWS totals approximately $217.00 per month and $2,604.00 per year, while Azure totals approximately $492.91 per month and $5,914.92 per year for the same three-service architecture.

Azure remains a strong option for organizations already invested in Microsoft technologies, especially when Azure Hybrid Benefit and Reserved Instances are available. However, for this small web application comparison, AWS is more cost-effective based on the selected calculator estimates.

12. GitHub Repository Contents

Folder/File	Purpose
README.md	Main project documentation for GitHub
Report/ AWS_vs_Azure_Improved_Report.pdf	Submission-ready PDF report
Report/ AWS_vs_Azure_Improved_Report.docx	Editable report file
Spreadsheet/Regional_Cost_Analysis.xlsx	Detailed monthly, annual, and percentage calculations
Screenshots/	Calculator screenshots and evidence
Calculator-Notes.md	Pricing assumptions, methodology, and rubric checklist

References

AWS Pricing Calculator: https://calculator.aws/

Azure Pricing Calculator: https://azure.microsoft.com/en-us/pricing/calculator/